06 Accountable, not fluent: harris-mcp, an open Model Context Protocol server for LLM-assisted reading of Harris matrices

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Large language models are entering archaeological practice mostly as text in, text out: the model is shown a screenshot of a Harris matrix, or a prose summary of one, and asked to reason over it. This discards the structure that makes the matrix useful — it is a partial order, not a picture — and leaves no way to check what the model actually did.

Harris-mcp is an Apache-2.0 Model Context Protocol server that exposes Harris matrices to any MCP client as typed, queryable, editable graphs. Deterministic graph work stays on the server (layers, ancestry, paths, contemporaneity, phases, diagnostics); interpretation stays with the model. Every model-driven edit lands on a per-matrix changelog with note and author; history, diff and revert are tools, so the interpretive trail is reviewable. It reads an open JSON pivot, CSV, XLSX, T. S. Dye's Frictionless hm data packages and Harris Matrix Composer's HMCX and CSV exports.

On limits, we test what can be tested. When two teams record the same deposits, which units are the same? propose_reconciliation matches units across matrices by description and stratigraphic agreement, refuses matches that contradict the sequence, and writes nothing until a person or the model asserts it. We score it on a real matrix — Çatalhöyük Buildings 1 and 5, 705 units — re-recorded by two simulated teams with their own numbering, noisy descriptions, missed relations, split and lumped units. Across trenches, description alone reaches F1 0.32 and leaves ~530 units in contradiction; period and type lift it to 0.95, the cycle check removes every contradiction, and structure is what survives noise (0.83 vs 0.70 at 20%).

What this does not show: descriptions are synthetic, the site is one, heavy noise still defeats it, and the model itself is not yet benchmarked, though the same harness can score an LLM, or an archaeologist, triaging the proposals. The binding constraint on AI in archaeology is not fluency. It is accountability.